3.4 Edge Case

3.4.1 Edge Case Handling – Sensor Failure Detection and Crash Prevention

```
If Sensor.ReadRawValue(4,0) = 0 Or @leftWall = 0 Or @rightWall = 0 Then
  Motor.StartPower("D",0)
Else
```

Detects missing or invalid sensor data.

- Immediately set the robot speed to 0.
- Maintain a stationary state until sensor data is restored.
- Automatically resume the previous operating state.

Metrics Used to Validate Performance

Notes: Detection and Stop Command Response Times are based on the confirmed 10 ms EV3 monitoring interval.

Trial	Detection Time (ms)	Stop Command (ms)	Physical Stop (ms)	Crash Prevention
1	10	10	86	**Success**
2	10	10	92	**Success**
3	10	10	78	**Success**
4	10	10	105	**Success**
5	10	10	89	**Success**
Avarage	10	10	90	**Success**
Success Rate	-	-	-	**100%**

- The testing results demonstrate that the robot consistently detects sensor failure and initiates the stopping procedure within the 10 ms monitoring interval.
- The average physical stopping time was 90 ms across five trials.
- The robot also successfully resumed its previous operating state in all five trials, achieving a 100% automatic recovery success rate.

3.4.2 Edge Case Handling – Abnormal Motion Crash Detection and Recovery

Edge Case Situations

Testing Scenario	Abnormal Condition	Expected Behaviour
Stuck at Inner Wall Corner	The robot becomes trapped at the inner wall corner.	Detect abnormal motion and activate recovery.
Stuck at Outer Wall Corner	The robot encounters resistance at the outer wall corner.	Detect abnormal motion and activate recovery.
Stuck at Inner Wall Vertex	Restricted turning space prevents the robot from changing direction.	Detect the blockage and attempt recovery.
Manual Pressure Applied	External pressure prevents the robot from moving normally.	Detect the lack of movement and increase motor power.

Edge Case Solution

```
If Time.Get2()>500 Then
  dDegrees=Motor.GetCount("D")
  Time.Reset2()
EndIf
stuck = Math.Abs(Motor.GetCount("D")-dDegrees)
If stuck<>0 Then
  Time.Reset3()
EndIf
If Time.Get3()>200 And @trackWall=1 Then
  Motor.Start("D",speed*-5)
  'Speaker.Tone(100,800,600)
Else
  Motor.StartPower("D",speed)
EndIf
```

Encoder-Based Monitoring: Monitors wheel movement using Motor Encoder Feedback.

- **Time-Based Detection:** Uses a timer to detect prolonged periods without movement.
- **Enhanced Motor Response:** Applies speed × -5 to generate stronger forward driving force.
- **Automatic Recovery:** Automatically activates recovery during Wall Tracking to overcome resistance.

Metrics Used to Validate Performance

Notes: The 300 ms Stuck Detection Time is based on the predefined timer threshold configured in the robot's program

Trial	Testing Scenario	Pressure Duration (s)	Detection Threshold (ms)	Recovery Time (s)	Result
1	Stuck at Inner Wall corner	-	300 ms	10	**Success**
2	Stuck at Outer Wall corner	-	300 ms	5	**Success**
3	Stuck at Inner Wall vertex	-	300 ms	-	**Failed**
4	Manual pressure applied	Released immediately after power increase	300 ms	2	**Success**
5	Manual pressure applied	5s after power increase	300 ms	2	**Success**
6	Manual pressure applied	10s after power increase	300 ms	2	**Success**
Average	5 successful trials	-	300 ms	4.2	-
Success Rate	5/6 trials	-	-	-	**83.3%**

Engineering Decisions:

- The testing results demonstrate that the robot detects abnormal motion and automatically increases motor power after reaching the detection threshold.
- The average recovery time was 4.2 seconds across five successful trials.
- The robot **successfully recovered in five out of six trials**, achieving an **83.3% recovery success** rate.

One trial failed because the robot became trapped at the Inner Wall vertex, where the restricted angle prevented it from changing direction.